

CYBERNOVA TECHNOLOGIES
ACTIVE DIRECTORY CAPSTONE PROJECT BRIEF

1. PROJECT TITLE

Enterprise Identity and Access Management (IAM) Deployment Using Active Directory

2. OVERVIEW

This capstone project provides students with a practical, real-world experience in designing and implementing a complete Active Directory (AD) environment for an enterprise organization.

Students will act as **Junior Systems Administrators** tasked with building and configuring a secure, structured, and functional domain environment for a fictional company, CyberNova Technologies.

The focus of this project is on:

- Active Directory deployment and configuration
- User and group management
- Organizational Unit (OU) design
- Group Policy Object (GPO) implementation
- Access control and resource sharing
- Security hardening and system administration best practices

3. SCENARIO

CyberNova Technologies is a growing organization with multiple departments and increasing needs for centralized identity management and secure access control.

The organization requires:

- Centralized authentication and authorization
- Structured department-based access control

- Secure file sharing across departments
- Enforcement of IT policies across all systems
- Proper delegation of administrative responsibilities

Students are required to design and implement a solution that meets these organizational needs.

4. PROJECT OBJECTIVES

By the end of this project, students should be able to:

- Deploy and configure an Active Directory Domain Services environment
- Create and manage users, groups, and organizational units
- Implement Group Policy Objects (GPOs) for security and administration
- Design and enforce role-based access control (RBAC)
- Configure secure file sharing and permissions
- Apply basic system hardening techniques
- Document technical implementations professionally

5. SYSTEM REQUIREMENTS

Students are required to set up a virtual lab environment consisting of:

- 1 Windows Server 2019/2022 (Domain Controller)
- 2–3 Windows 10/11 Client Machines

(VirtualBox or VMware may be used)

6. DOMAIN CONFIGURATION

- Domain Name: **cybernova.local**
- Install and configure Active Directory Domain Services (AD DS)
- Promote the server to a Domain Controller

7. ORGANIZATIONAL UNIT (OU) STRUCTURE

Students must design and implement the following structure:

- IT Department
- HR Department
- Finance Department
- Remote Users

Students are expected to justify their OU design in their final report.

8. USER AND GROUP MANAGEMENT

Students must:

- Create a minimum of 8–12 user accounts across all departments
- Create and configure security groups such as:
 - IT_Admins
 - HR_Staff
 - Finance_Staff
 - Remote_Users

All users must be assigned appropriate group memberships based on their roles.

9. GROUP POLICY OBJECTS (GPOs)

Students must configure and apply the following policies:

- Password policy enforcement (minimum length and complexity)
- Account lockout policies after multiple failed login attempts
- Restriction of Control Panel access for standard users
- Restriction of command-line tools for non-administrative users
- Automatic screen lock after inactivity

- Desktop or organizational policy settings

10. FILE SHARING AND PERMISSIONS

Students must create and configure secure shared folders:

- IT Share
- HR Share
- Finance Share

Each folder must be configured with:

- Appropriate NTFS permissions
- Share permissions
- Role-based access control (RBAC)

Access must strictly follow the principle of least privilege.

11. SECURITY HARDENING REQUIREMENTS

Students must apply basic system security practices including:

- Disabling unnecessary services
- Enabling Windows Firewall rules
- Configuring auditing policies (logon and account activity tracking)
- Applying secure administrative practices
- Documenting backup and recovery procedures

12. PROJECT DELIVERABLES

Each student or group must submit the following:

1. Technical Report

- System architecture overview
- Step-by-step implementation process with screenshots

- OU and domain design explanation
- GPO configuration details

2. Network Diagram

- Visual representation of the AD environment and connected systems

3. Security Documentation

- Applied security policies
- Access control design
- Justification of security decisions

4. Operational Guide

- User creation process
- Group management procedures
- Password reset process
- Resource access management

13. ASSESSMENT CRITERIA (TOTAL: 100%)

- Active Directory Setup & Configuration – 20%
- OU Design & Structure – 15%
- User & Group Management – 15%
- Group Policy Configuration – 20%
- File Sharing & Permissions – 10%
- Security Hardening – 10%
- Documentation Quality – 10%

14. PROJECT TIMELINE

- **Project Start Date:** Immediately upon assignment
- **Submission Deadline:** 14th June, 2026

Late submissions may attract penalties unless otherwise approved.

15. GENERAL INSTRUCTIONS

- This is an individual
- Students must ensure originality in their implementation and documentation
- Screenshots must be included as evidence of all configurations
- Professional documentation standards must be followed
- All work must be submitted in Microsoft Word or PDF format

16. FINAL NOTE

This project is designed to simulate a real enterprise environment and prepare students for careers in systems administration, cybersecurity, and IT infrastructure management.

Students are encouraged to focus on **accuracy, structure, and professionalism** in their work.